

30 *Use of the Data Booklet is relevant to this question.*

A student noted the relevant standard electrode potentials and correctly deduced that bubbling $\text{O}_2(\text{g})$ through an acidic solution of $\text{HBr}(\text{aq})$ should oxidise it to $\text{Br}_2(\text{aq})$.

On bubbling air through an acidic solution of $\text{HBr}(\text{aq})$ in a test tube, the student observed no effect.

What could be reasons for this?

- 1 The oxygen pressure was smaller than 1.0 atmosphere.
- 2 The activation energy for the reaction is high.
- 3 The $\text{Br}^-(\text{aq})$ concentration is much smaller than 1.0 mol dm^{-3} .

A 2 only

B 3 only

C 1 and 2 only

D 1, 2 and 3